Zackery Beyer

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Education

University of Missouri — Columbia, MO
Bachelor of Engineering in Computer Science, Minor in Mathematics — *Aug. 2023 – Dec. 2026*
– Certificate: Web and Mobile Applications Development — **GPA: 3.95/4.00**

Experience

Garmin International — May 2026 – Aug. 2026
Software Engineering Intern, Automated Test Equipment (ATE) Software — *Olathe, KS*
– **Traced and fixed the main bottleneck** on a high-traffic ATE test-report platform, restructuring the SQL query patterns behind a large-report view to cut load time **from 33s to 1.6s (∼95%)**; replaced a full yield scan with a targeted step-name lookup that dropped query volume **80%+**.
– **Built a fault-tolerant PC provisioning tool** that **replaced a manual multi-hour setup**: a normalized **SQL Server** schema defines install steps as data (new software added without code changes), driven by a **PyQt5** client, with per-step retry/skip and **checkpointed reboot-and-resume** so setups finish unattended across restarts.
– **Wrote two instrument drivers in C** against vendor SDKs over **NI-VISA/NI-MAX**, plus a GUI for each, so automated test routines drive the hardware directly instead of by hand; **peer-reviewed and documented** for team handoff.
– **Owned each project end to end** on an established team (design, code review, testing, deployment, support): shipped a batch comparison view, **used daily**, replacing a manual serial-by-serial lookup. **Received a full-time return offer.**

DataAnnotation — 2026 – Present
Software Engineering Contractor, AI Coding Evaluation — *Remote*
– **Critically evaluate AI-generated code** from frontier LLM coding agents for correctness, edge cases, security, maintainability, and alignment with stated requirements, **accountable for the final verdict** and documenting the failure mode behind each rejection.
– **Design adversarial coding and Docker tasks** against real open-source codebases (e.g. `pandas`) to surface where AI development tools break down, and **iteratively steer agents** to correct output **with human review as the control**.

Projects

Homelab | *Raspberry Pi 5, Kubernetes (k3s), Go, Docker, Prometheus, Grafana, Linux*
– **Run a self-healing k3s cluster** on a Raspberry Pi 5, migrated from version-controlled Docker Compose to declarative Kubernetes, with a custom **Go** service handling **health checks, backups, and automated recovery** with no operator in the loop.
– **Instrumented the stack with Prometheus and Grafana** for service health and resource alerting, and **hardened access** with automated TLS, private Tailscale networking, and key-only SSH.

FaultInjector | *.NET 10, C#, ASP.NET Core, Docker*
– Built a **Dockerized fault-injection reverse proxy**: custom **ASP.NET Core middleware** injects controlled latency, errors, and timeouts into live API traffic to **test how services behave when dependencies fail** (chaos engineering).
– Added a **control-plane REST API** to swap fault rules at runtime, so failure experiments run with **no restart or redeploy**.

MU Planner | *Next.js, React, Python, GitHub Actions*
– Built **Python ETL scrapers** merging the course catalog and RateMyProfessors into a unified schema, **refreshed automatically on scheduled GitHub Actions**, and shipped the Next.js frontend to **100+ Mizzou students**.

Boeing Defense Flight Simulator | *Senior Capstone, 5-person team*
– **Own the geospatial data pipeline** (QGIS → BlenderGIS → Blender) for simulator-ready airport models, holding **positional tolerance within a few feet** against a formal client requirements doc.

Technical Skills

Languages: Python, Go, C#, C++, C, SQL, TypeScript, JavaScript
Engineering Practice: Secure coding, code review, unit/integration testing, technical documentation, CI/CD (GitHub Actions), Agile delivery, Git
AI-Assisted Development: LLM coding agents (Claude Code, Copilot) for drafting, refactoring, testing, and documentation; **validation of AI-generated output**
Systems & Infrastructure: Docker, Kubernetes (k3s), Linux, microservices and REST APIs, resiliency and fault-injection testing, Prometheus, Grafana, monitoring and alerting
Data: PostgreSQL, SQL Server, MongoDB, schema design, query optimization, ETL pipelines